

Final Project Proposal

Problem Statement & Motivation:

- **What real-world business problems am I addressing?**

- The real-world business problem I seek to address concerns the stock market and the time value of money. Individuals are being burned by making poor investment decisions regarding the stock market and not setting aside money for their future. Because of this, I would like to create an application or service that can help our generation identify the best methods for long-term financial success, whether it be Index funds, Bonds, Stocks, etc.

- **Who are my target users?**

- My target users are individuals aged 18+ who would like some sort of financial guidance or assistance to help prepare for their future.

- **Why is an Agentic AI solution appropriate here?**

- I find an Agentic AI solution to be appropriate here to accommodate users. Financial Advisors may cost money and can only schedule you based on their availability. My application seeks to be available 24/7 and to use real-time financial data to make structured decisions based on the query. We are moving into a world centered on Artificial Intelligence, and this could be a pivotal step in that direction.

Use Cases and Scenarios:

- **2–3 concrete user scenarios:**

- *Scenario A:*

- **Persona:** Evan Anderson, 18, fresh high school graduate
 - **Goal:** Figure out how to prepare for his future
 - **Context/Trigger:** Evan recently had a conversation with his parents discussing the time value of money and how saving for retirement is no longer enough.
 - **Scenario Action Steps:**
 - Evan does external research to identify apps that fit his goal.
 - He stumbles across [name to be determined] and signs up to be a user and asks the system what's the best way to ensure financial security long-term.
 - He receives a simplified version of investment information with a Disclaimer at the bottom informing the user that they are not responsible for the success or failure of any investment.
 - **Agent Tasks:**
 - Utilize APIs to retrieve financial market data (APY, Interest Rates, etc.) to build a simplified report in a pretty package for Evan.

- *Scenario B:*

- **Persona:** Joey Sassafras, 45, Options Day Trader
 - **Goal:** Identify profitable Options for the day

- **Context/Trigger:** Joey, age 45, is on a terrible losing streak. He's determined that his current prediction methods are dated and wishes to "get with the times".
- **Scenario Action Steps:**
 - Joey does research to find what the kids are using to chase the success they have experienced
 - He finds [name to be determined] and signs up to be a user
 - He fills out information about his level of understanding and requests historical information about the options available
 - He receives a list of options ranked by % for the pulls/puts to be successful based on the information we've seen beforehand.
- **Agent Tasks:**
 - Utilize Financial Data APIs to display Trends of the options
 - Perform Analysis to diagnose trend changes
 - Compute a profitability score based on diagnostic

Initial System Design:

- **Agents:**
 - A Planning Agent for that can identify the needed agent(s)
 - A Data fetcher agent to pull historical financial data
 - An Analyst Agent that can calculate the math for the financial data
 - A Disclaimer Agent that includes the necessary disclaimers for liability purposes

- An Educator/summarizer Agent to simplify the terminology for newer/younger users.

- **Tools/APIs:**

- Yfinance API for stock prices and historical data
- FRED API for interest rates and economic data
- Google Opal for building the demo and connecting agents

- **Feasibility & Risks:**

- **Legality:**

- Since I am not a licensed/professional advisor, I need to indicate that outputs are meant to be used for educational purposes instead of investment advice.

- **Hallucination:**

- AI has the tendency to be incorrect with math, I found to help mitigate this, I can force it to use python to calculate figures, then use the AI to display the result

- **Timeline:**

- Week 1: Research of APIs & how to build a cross functional system
- Week 2-3: Architecture & API setup. Get the "Researcher" agent to pull real data.
- Week 3-4: Logic & Analysis. Build the "Analyst" agent and test math accuracy.
- Week 4-5: Persona Tuning. Ensure new users get simple charts, and users like Joey get "the pulls/puts" data he wants.

- Week 5-8: UI Development & Final Testing.